

Stable Scores, Failed Retrieval: Diagnosing RAG Evaluation in Low-Resource Indic Languages

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Abstract

Automatic metrics stand in for native-speaker evaluation in low-resource retrieval-augmented generation (RAG), so their validity determines what can be concluded about these systems. We test whether common metrics detect retrieval and grounding failure in one fixed RAG pipeline across ten Indic languages. They do not merely miss the failure in Santali; they invert it. Hit@5 is 0.011, yet raw XLM-R BERTScore is the highest in the set at 0.902, changes little when retrieved evidence is removed, and assigns nearly the same score to matched and unrelated references. The cause is coverage: the scorer maps 39% of Santali tokens to <unk>, and a scorer whose vocabulary includes Ol Chiki restores discrimination while LaBSE does not. Retrieval fails for the same reason. The gold article is indexed for 80% of Santali queries, but the retriever shares XLM-R’s tokenizer and collapses before reranking. Judge-based faithfulness and context precision, computed by a model with a different tokenizer, flag the failure independently. A metric that cannot represent a script does not become noisy; it becomes constant and high. We argue that metric validity in low-resource languages must be established per language, and that evidence removal, reference shuffling, and the scorer’s <unk> rate can be important additions.

1 Introduction

Retrieval-augmented generation (RAG) is designed to condition generation on external evidence (Lewis et al., 2020). Evaluating a RAG system therefore requires separating at least three questions: whether the system retrieved the expected evidence, whether the answer resembles a correct reference, and whether the answer is supported by the retrieved context. These questions are related, but they are not interchangeable. A model

can produce a plausible answer from parametric knowledge after retrieval fails, and an answer can resemble a reference while remaining unsupported by the supplied evidence.

The stakes are practical. Automatic scores are what multilingual benchmarks, model cards, and internal deployment reviews report, and they are relied on most heavily for exactly the languages where no one on the team can read the output. A metric that fails by producing low scores is caught; a metric that fails by producing high, stable scores is not. This paper documents the second kind of failure: in the language where our pipeline retrieves almost nothing, the most widely reported answer-quality metric returns its highest value, and nothing in that number signals a problem.

Standard automatic metrics observe different parts of this process. Hit@K measures retrieval of an expected source. Token F1 and BERTScore (Zhang et al., 2019) compare a generated answer with a reference answer. RAGAS (Es et al., 2024) uses an LLM judge to assess constructs such as faithfulness, context precision, and answer relevancy. These metrics scale without language-specific annotation, which is useful when native-speaker evaluation is unavailable. Weak coverage of a language or script, however, can make their scores difficult to interpret without independent diagnostics.

We study this problem using ten Indic languages that vary in corpus size, script, and resource level. All languages use the same retrieval, reranking, and generation pipeline. Our central question is: when retrieval fails in a low-resource language, do common RAG evaluation metrics detect the failure, and what diagnostics explain disagreement among them?

We make three contributions. First, we show a concrete failure of metric interpretation: raw XLM-R BERTScore remains nearly unchanged when evidence is removed and is highest for Santali despite

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near-zero retrieval. Second, we test two alternatives left open by that intervention. Shuffled references and alternative scorers distinguish grounding insensitivity from semantic discrimination failure, while index presence, pre-reranking retrieval, and embedding diagnostics localize the Santali retrieval failure. Third, we compare these signals with RAGAS judge metrics under one fixed protocol and report metric-specific validity. We recommend that evaluation in undercovered languages test retrieval, evidence sensitivity, scorer discrimination, and judge validity separately.

2 Related Work

RAG evaluation. The original RAG formulation evaluates both retrieval and downstream task performance (Lewis et al., 2020). Reference-based metrics such as token F1 and BERTScore evaluate the generated answer, while RAGAS separates context precision, faithfulness, and answer relevancy (Zhang et al., 2019; Es et al., 2024). We test whether each signal responds to the property it is used to support, especially when retrieval is poor.

Multilingual judges and RAG evaluation. LLM-as-a-judge methods make open-ended evaluation scalable, but judge behavior can vary across prompts, models, and languages. Prior work documents position and self-preference biases (Zheng et al., 2023; Shi et al., 2025; Wataoka et al., 2024), and multilingual judge reliability can degrade in lower-resource settings (Fu and Liu, 2025). MEMERAG extends RAG meta-evaluation beyond English using native-language questions and expert judgments (Cruz Blandón et al., 2025). We study a complementary question: how standard metrics behave under controlled evidence removal in low-resource Indic languages.

Indic language evaluation and representation coverage. IndicGenBench and IndicParam document uneven model performance across Indic languages (Singh et al., 2024; Maheshwari et al., 2025). Cross-lingual transfer also depends on factors beyond corpus size, including language relationships and script sharing (Bafna et al., 2023). Tokenizer choice can materially affect multilingual model performance (Rust et al., 2021; Ali et al., 2024), and BERT-based evaluation metrics can lose robustness when inputs diverge from the representation space of the scorer (Vu et al., 2022). These findings motivate direct tests of tokenizer cover-

age and representation geometry instead of treating resource tier as an explanation by itself.

Validating evaluation metrics by perturbation.

Controlled perturbations test whether an automatic metric responds to the property it is intended to measure (Sai et al., 2021; Kaster et al., 2021). Fine-grained analyses also show that BERT-based metrics respond unevenly to different linguistic factors (Hanna and Bojar, 2021). Contextual embedding spaces are anisotropic (Ethayarajh, 2019), which contributes to high raw cosine similarities and motivates baseline correction when interpreting embedding similarity. XLM-R was trained on CC-100 (Conneau et al., 2020); its published 100-language inventory does not include Santali. BGE-M3 uses the XLM-R tokenizer (Chen et al., 2024). Our evidence-removal and shuffled-reference tests apply this perturbation logic to grounding in a setting where scorer coverage varies by script.

3 Methodology

3.1 Testbed and Retrieval Pipeline

We use 1,000 queries per language from ai4bharat/Indic-Rag-Suite. Each example provides a question, gold answer, and source Wikipedia article. We evaluate Bengali, Tamil, Telugu, Hindi, Urdu, Marathi, Gujarati, Nepali, Santali, and Maithili. Resource labels are descriptive: Marathi, Gujarati, and Nepali are labeled low-resource and Maithili and Santali extremely low-resource following IndicParam where available (Maheshwari et al., 2025).

For each language, we build a language-specific index over passages from the Cohere multilingual Wikipedia corpus, CohereLabs/wikipedia-2023-11-embed-multilingual-v3. Passages and queries are embedded with BAAI/bge-m3 (Chen et al., 2024). BGE-M3 produces 1,024-dimensional vectors, which we L2-normalize and store in a FAISS IndexFlatIP (Johnson et al., 2019); inner product therefore implements cosine ranking. We retrieve 25 passages, rerank them with Jina reranker v3 (Wang et al., 2025), and pass the top five to llama3.1:8b through Ollama (Grattafiori et al., 2024). The prompt asks the model to answer in the query language and cite used passages. Generation uses temperature 0. Gold answers and source passages are used only for evaluation or for the explicitly defined gold-context intervention.

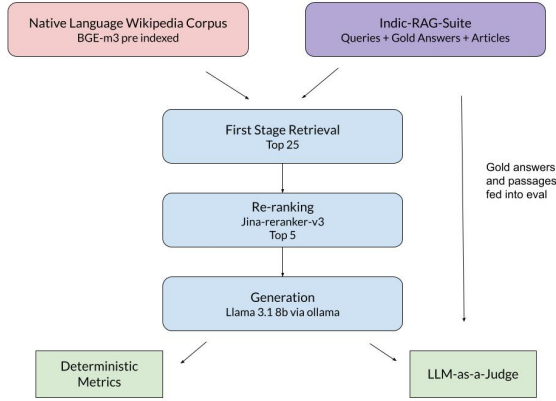


Figure 1: RAG pipeline used for all ten languages.

3.2 Evaluation Metrics

Retrieval. Hit@5 records whether the gold Wikipedia article appears among the five final reranked passages. Article titles are normalized with NFC, Unicode case-folding, Unicode punctuation removal, and whitespace normalization before matching.

Reference-based answer metrics. Token F1 uses the same Unicode normalization and computes multiset F1 over whitespace-delimited tokens. BERTScore F1 is computed with xlm-roberta-large (Zhang et al., 2019). We report raw, unrescaled BERTScore because language-specific rescaling baselines are not available for the full language set. The semantic-discrimination control also uses ai4bharat/IndicBERTv2-MLM-only (Doddapaneni et al., 2023) with BERTScore and sentence-transformers/LaBSE (Feng et al., 2022) with sentence-level cosine similarity. LaBSE scores are a different quantity and are not compared in absolute value with BERTScore.

Judge-based metrics. We use RAGAS faithfulness, context precision, and Answer Relevancy (Es et al., 2024). Faithfulness tests whether answer claims are supported by supplied contexts. Context precision judges whether relevant contexts are ranked near the top. Standard Answer Relevancy generates three reverse questions per response (strictness=3) and compares them with the original query using embeddings. Faithfulness and context precision are defined only for context-bearing conditions, while Answer Relevancy is defined for all three conditions. We treat all judge scores as diagnostic signals rather than human-validated ground truth.

3.3 Diagnostic Interventions

Evidence sensitivity. We generate answers under three conditions for the same 1,000 query IDs per language: *gold*, where context comes from the gold article; *retrieved*, the original top-five reranked passages; and *no-context*, where no passages are supplied. The retrieved condition is the baseline output; gold and no-context add 20,000 generations. This intervention tests whether an answer metric changes when evidence availability changes. It does not assume that a no-context answer is semantically incorrect, since a generator may answer from parametric knowledge.

We quantify uncertainty with paired bootstrap intervals over query-level metric differences. Per-language intervals use 10,000 resamples. For pooled results, we use a two-level cluster bootstrap that resamples languages and then queries within each sampled language.

Semantic discrimination. To test semantic discrimination independently of context availability, each generated answer is scored against its matched gold answer and against gold answers from five fixed seeded derangements within the same language. No example is paired with itself. We report the matched-minus-shuffled gap for XLM-R BERTScore, IndicBERTv2 BERTScore, and LaBSE cosine, together with each scorer tokenizer’s <unk> rate.

Retrieval localization. Hit@5 can fail because the gold article is absent from the index, because dense retrieval misses it, or because reranking removes it. We therefore audit gold article presence, dense Hit@25 before reranking, and final Hit@5. We also measure mean pairwise cosine among query embeddings, cosine to the language query centroid, and the BGE-M3 tokenizer <unk> rate.

Judge validity. RAGAS uses the same fixed 200 query IDs per language and condition. We report metric-specific valid sample counts rather than imputing invalid outputs, and raw judge responses and errors are retained for audit. The structural output audit is reported in the appendix because it checks form, including missing or invalid citations, rather than semantic grounding.

3.4 Reproducibility

Code, prompts, configurations, evaluation scripts, and provenance utilities are avail-

able at <https://github.com/jasgust/low-resource-rag-eval>. The RAGAS judge is Qwen/Qwen2.5-7B-Instruct (Yang et al., 2024), revision prefix a09a35458c70, served in bfloat16 with vLLM 0.29, temperature 0, and a 16,384-token serving limit. Standard Answer Relevancy uses BAAI/bge-m3 revision prefix 5617a9f61b02 for embedding similarity and strictness=3. Judge calls allow at most two retries; the same 200 sample keys are used across conditions, and metric-specific invalid outputs are left missing. The repository pins the full model revisions, RAGAS and Python dependencies, input hashes, scorer settings, shuffle seeds, Git commits, and checksums for diagnostic artifacts.

The archived generation run records the Ollama model tag llama3.1:8b and temperature 0 but did not set an explicit generation seed. Exact scorer layers and other scorer settings are specified in the released scripts rather than reconstructed from prose.

4 Results

4.1 Metric Disagreement Identifies a Stress Case

Table 1 establishes the baseline disagreement that motivates the diagnostics. Gujarati has the highest Hit@5 (0.917) and token F1 (0.205). Santali has the lowest Hit@5 (0.011) and token F1 (0.066), yet the highest raw BERTScore (0.902). Resource tier alone does not explain the pattern: Maithili is also labeled extremely low-resource but reaches Hit@5 of 0.522, and Gujarati is labeled low-resource but retrieves best.

The Santali contrast is the paper’s central puzzle. Retrieval indicates that the expected evidence is almost never in the final context, while the reference-similarity metric gives Santali the highest score in the set. The remaining experiments test that disagreement without assuming that either signal is ground truth.

4.2 Evidence Removal Leaves BERTScore Nearly Stable

Table 2 summarizes the context intervention; the full language-level table is in Appendix A. Token F1 is responsive to context removal. Its macro mean falls from 0.158 with retrieved context to 0.058 with no context, and no-context F1 is less than half the retrieved value for nine of ten languages. Santali is the exception, with 0.066 re-

trieved and 0.083 no-context. This reversal is consistent with the retrieval failure: irrelevant retrieved context can steer generation toward unrelated vocabulary, while removing that context avoids the interference. Token F1 is therefore a context-sensitive lexical diagnostic here, not a grounding measure.

BERTScore changes much less. The macro mean is 0.852 with gold context, 0.850 with retrieved context, and 0.848 with none. The pooled gold-minus-no-context BERTScore difference is 0.0040, with a 95% cluster-bootstrap interval of [0.0019, 0.0058]. The corresponding token-F1 difference is 0.0719 [0.0544, 0.0918]. For Santali, BERTScore is 0.905 with gold context, 0.902 with retrieved context, and 0.907 with no context. Its gold-minus-no-context difference is -0.0019 [-0.0031, -0.0008]. Thus raw BERTScore is weakly sensitive to evidence availability in this setting and cannot by itself support a grounding claim.

This intervention establishes evidence insensitivity, not general semantic failure. A no-context answer may still be correct. We therefore test semantic discrimination directly.

4.3 Santali Semantic Discrimination Depends on the Scorer

Table 3 reports matched-minus-shuffled gaps. Under XLM-R, Santali’s retrieved-condition gap is 0.0015: matched references score 0.9022 and shuffled references 0.9008. The other nine languages average a gap of 0.0219. Raw XLM-R gaps are small across languages, consistent with the anisotropy of contextual embedding spaces (Ethayarajh, 2019). Because a Santali baseline for BERTScore rescaling is unavailable, the shuffled-reference condition provides an in-domain discrimination check. Within XLM-R, Santali’s retrieved gap is 7% of the other-language mean. The corresponding ratio is 47% for IndicBERTv2 on retrieved answers and approximately 100% under gold and no-context. The other-language XLM-R gap of 0.0219 is also about five times the pooled gold-minus-no-context change of 0.0040, so the evidence intervention lies within a range that the scorer can resolve. The XLM-R tokenizer maps 38.7% of Santali answer tokens to <unk>.

With IndicBERTv2, Santali’s <unk> rate is 0.08%, and the gold and no-context discrimination gaps (0.0794 and 0.0589) match the other-language averages (0.0796 and 0.0581). The retrieved-condition gap is lower (0.0426 versus 0.0900),

Lang	Tier	Hit@5	Token F1	XLM-R BERTScore
Bengali (bn)	High	0.796	0.177	0.846
Tamil (ta)	High	0.805	0.187	0.855
Telugu (te)	High	0.534	0.147	0.850
Hindi (hi)	High	0.817	0.186	0.852
Urdu (ur)	Mid	0.511	0.175	0.824
Marathi (mr)	Low	0.525	0.140	0.838
Gujarati (gu)	Low	0.917	0.205	0.859
Nepali (ne)	Low	0.743	0.152	0.841
Santali (sat)	X-low	0.011	0.066	0.902
Maithili (mai)	X-low	0.522	0.143	0.836

Table 1: Retrieved-condition baseline results over 1,000 queries per language. BERTScore is raw and unrescaled. Bold marks column maxima.

Summary	Token F1			BERTScore		
	Gold	Retr.	No-ctx	Gold	Retr.	No-ctx
Macro mean	0.153	0.158	0.058	0.852	0.850	0.848
Santali	0.144	0.066	0.083	0.905	0.902	0.907

Table 2: Context-ablation summary. Full language-level results are in Table 6.

which is consistent with poorer retrieved answers after retrieval failure. LaBSE does not recover the distinction: its Santali gap remains far below its other-language average and its tokenizer maps 61.3% of Santali tokens to <unk>. The result is scorer-dependent and is consistent with representation coverage affecting semantic discrimination.

4.4 Santali Fails in Dense Retrieval Before Reranking

The retrieval diagnostics rule out simple corpus absence as the primary explanation. The gold article is present for 796 of 1,000 Santali queries. Only 17 reach the dense top 25 and 11 reach the final top five. Of those 17 dense successes, reranking removes six. The major loss therefore occurs before reranking.

Santali’s embedding geometry is also anomalous. Its mean pairwise query-embedding cosine is 0.960, compared with 0.282 to 0.314 for the other nine languages. Its mean cosine to the query centroid is 0.980, compared with 0.531 to 0.560. BGE-M3 maps 47.85% of Santali query tokens to <unk>, while the other languages are effectively at zero. These measurements localize an extreme representation problem in this model stack; they do not establish that tokenizer coverage alone causes the retrieval failure.

4.5 Judge Metrics Expose Different Constructs

Faithfulness and context precision provide evidence-sensitive signals. For nine of ten languages, their retrieved-context means exceed their gold-context means. Santali is the sole exception: faithfulness falls from 0.339 to 0.292, and context precision from 0.357 to 0.148. The direction for the other nine languages is plausible under our intervention: the gold condition supplies passages from the source article rather than an oracle answer span, whereas retrieved passages are explicitly selected for the query and reranked. For those languages, retrieved context can therefore contain more judge-visible support than the sampled gold-article context. Santali reverses the pattern because its retrieval pool collapses before reranking.

Answer Relevancy measures whether the response addresses the question, not whether it is grounded in supplied evidence. Its macro mean rises from 0.379 with gold context to 0.433 with retrieved context and 0.552 with no context. Unsupported answers can therefore remain topically related to the query. Santali’s Answer Relevancy is nearly constant and near zero (0.031/0.038/0.039). Inspection of the reverse-question outputs shows that the judge frequently produces generic English questions about the OI Chiki text or marks the answer noncommittal. The similarity stage also uses BGE-M3, whose tokenizer maps 47.85% of Santali

Scorer	Santali <unk>	Retrieved		Gold		No-ctx	
		sat	other 9	sat	other 9	sat	other 9
XLM-R BERTScore	38.7%	0.0015	0.0219	0.0045	0.0191	0.0018	0.0107
IndicBERTv2 BERTScore	0.08%	0.0426	0.0900	0.0794	0.0796	0.0589	0.0581
LaBSE cosine	61.3%	0.0140	0.1923	0.0364	0.1711	0.0229	0.1162

Table 3: Matched-minus-shuffled score gap over five seeded derangements. Values compare Santali with the mean of the other nine languages.

Diagnostic	Santali	Other nine
Gold title in index, queries	796/1000	–
Dense Hit@25	1.7%	54.9–94.9%
Final Hit@5	1.1%	51.1–91.7%
Mean pairwise query cosine	0.960	0.282–0.314
Mean cosine to centroid	0.980	0.531–0.560
Query <unk> rate	47.85%	0%

Table 4: Santali retrieval and representation diagnostics. Full language-level results are in Appendix A.

Signal	Gold	Retr.	No-ctx
Santali faithfulness	0.339	0.292	–
Santali context precision	0.357	0.148	–
Answer Relevancy, macro	0.379	0.433	0.552
Answer Relevancy, Santali	0.031	0.038	0.039

Table 5: Judge diagnostics. Full scores and valid counts are in Table 9.

query tokens to <unk>, so both the reverse-question generation and embedding-similarity stages have language coverage problems. We therefore do not interpret Santali Answer Relevancy as a calibrated cross-language quality score.

Validity is high but not uniform, so means are not computed over an assumed $n = 200$ in every cell. Faithfulness is valid for 1,977/2,000 gold and 1,986/2,000 retrieved examples; context precision for 1,877/2,000 and 1,860/2,000; Answer Relevancy for 1,988/2,000 gold, 1,991/2,000 retrieved, and 1,993/2,000 no-context examples. The appendix reports each cell’s effective n .

4.6 Supporting Structural Audit

The deterministic audit finds no empty or near-empty baseline answers and low truncation across languages. Citation omission is the main form-level failure, from 1.0% in Bengali to 36.7% in Santali. Because citation presence does not show that a cited passage supports a claim, we treat this audit as supporting output diagnostics and report the full table in Appendix A.

5 Discussion

5.1 What the Experiments Establish

The evidence supports two claims with different scope. First, across the ten languages, raw XLM-R BERTScore is weakly sensitive to evidence removal. Its pooled change is small relative to token F1, so a high BERTScore should not be interpreted as evidence that an answer is grounded in retrieved passages. Second, Santali exposes a stronger scorer-specific failure: XLM-R barely distinguishes matched from shuffled references. That result cannot be explained only by parametric knowledge because the reference is changed while the answer is held fixed. IndicBERTv2 restores discrimination under gold and no-context conditions, while LaBSE does not. Semantic discrimination therefore depends materially on the representation model and its coverage. These experiments do not validate IndicBERTv2 as a replacement metric.

5.2 Representation Coverage Is a Risk Factor, Not a Complete Cause

The retrieval failure is localized before reranking. Most Santali gold articles are present in the index, but almost none reach the dense top 25. The extreme query-embedding concentration and high <unk> rate provide direct measurements of poor representation coverage in this stack. XLM-R BERTScore shows a parallel coverage problem.

One mechanism consistent with these measurements is token collapse. BERTScore greedily aligns contextual token embeddings; in the implementation used here, <unk> positions remain in the

alignment. When a large fraction of Santali content on both candidate and reference sides maps to the same unknown token identity, those positions can impose a high similarity floor even when the underlying content differs. The 0.9008 shuffled-reference floor is consistent with this mechanism. BGE-M3 uses the same XLM-R tokenizer (Chen et al., 2024); Santali queries therefore lose lexical distinctions before encoding, consistent with their mean pairwise cosine of 0.960 and the narrow retrieved-title pool of 421 retrieved titles for 947 gold titles. IndicBERTv2 was trained for Indic languages including Santali in Ol Chiki (Doddapaneni et al., 2023) and maps only 0.08% of Santali tokens to <unk> in our data, consistent with its restored discrimination. Maithili uses Devanagari, a script represented by Hindi in XLM-R’s training inventory (Conneau et al., 2020), and reaches Hit@5 of 0.522 despite the smallest index. Script and tokenizer coverage are therefore a stronger explanation for the Santali-Maithili contrast than corpus size, although the experiments do not isolate coverage as the sole cause.

BGE-M3 and XLM-R are different encoders, so the experiments do not establish that the same internal mechanism causes both failures. Corpus size and resource tier are also insufficient explanations: Gujarati retrieves best despite its low-resource label, and Maithili substantially outperforms Santali with a smaller index.

Faithfulness and context precision identify the Santali retrieved-context problem, but Answer Relevancy does not provide a meaningful Santali comparison because both its reverse-question generation and BGE-M3 similarity stage have poor Ol Chiki coverage. Model coverage therefore matters at multiple points in an evaluation pipeline, including the retriever, semantic scorer, and judge.

5.3 Implications for Multilingual RAG Evaluation

Three properties of the Santali failure generalize beyond this stack.

First, undercoverage does not lower scores; it inflates and stabilizes them. A scorer that cannot represent a script returns a high similarity floor for any pair of strings in that script, and that floor does not move when evidence is removed or the reference is replaced. The worst-served language in an evaluation can therefore look like the best-served one, and a practitioner scanning a leaderboard has no way to tell from the number alone.

Second, the failure is correlated across components that are usually treated as independent checks. The retriever, the reference scorer, and the embedding model behind Answer Relevancy all share the XLM-R tokenizer here, so all three fail on the same 48% of tokens. Multi-metric evaluation guards against independent errors; it offers no protection when the metrics inherit one vocabulary. Faithfulness and context precision caught the Santali failure because the judge is a decoder model with a different tokenizer, not because the protocol had redundancy by design.

Third, the coverage variable is measurable before any experiment is run. A per-language <unk> rate for every model in the pipeline costs seconds and, in this study, separates the language where every metric is uninformative from the nine where they are not. This suggests a simple admissibility rule: a scorer, retriever, or judge embedder should be reported for a language only alongside its <unk> rate on that language’s text, and a rate above a few percent should be treated as disqualifying rather than as a caveat.

The evaluation protocol follows from these properties. Verify that expected evidence is in the index; report retrieval before answer scores; test any reference-similarity metric under evidence removal; test it against shuffled references; and report judge validity per language. Each check targets a failure that the others cannot see.

6 Conclusion

We set out to ask whether common RAG metrics detect retrieval failure in low-resource languages. In Santali they do not merely miss it; they invert it. The language in which the pipeline retrieves the expected article for one query in a hundred receives the highest answer-similarity score of the ten, that score does not move when the evidence is removed, and it barely moves when the reference is replaced with an unrelated one. A scorer whose vocabulary covers Ol Chiki recovers the distinction. The failure is a property of the scorer’s coverage, not of the language or the answers.

The broader implication concerns how evaluation infrastructure is built. The retriever, the reference scorer, and the embedder behind the judge’s relevance metric share one tokenizer, so they share one blind spot, and a protocol that combines them gains no independence from doing so. An evaluation stack cannot see what its own vocabulary

cannot represent, and when it cannot, it reports confidence rather than uncertainty. For languages outside the pretraining distribution of standard multilingual encoders, this means that a published score is a statement about the evaluator at least as much as about the system.

The practical response is to treat metric validity as something established per language rather than inherited from the metric. Two interventions with no annotation cost, evidence removal and reference shuffling, are sufficient to expose the failure, and the scorer’s <unk> rate predicts it before either is run. We argue these should accompany any cross-lingual report of raw similarity scores.

What remains open is how far the pattern extends. Native-speaker judgments would show whether the answers IndicBERTv2 now separates are in fact better or worse; other scripts absent from common vocabularies would show whether the inversion is general; and claim-level attribution would replace the citation-format audit with a grounding measure that does not depend on any encoder’s coverage.

Limitations

We evaluate one RAG stack: BGE-M3 retrieval, Jina reranker v3, llama3.1:8b generation, XLM-R-based BERTScore, and Qwen2.5-7B-Instruct for RAGAS judging. Results may differ with other retrievers, rerankers, generators, scorers, or judges. We also lack native-speaker human judgments of correctness and groundedness, so no automatic metric is treated as ground truth.

The semantic-discrimination experiment shows that scorer discrimination is recoverable for Santali, but it does not validate IndicBERTv2 as a general quality metric. Likewise, the near-zero Santali Answer Relevancy values should not be read as calibrated answer quality because both the reverse-question judge and BGE-M3 similarity stage have poor Ol Chiki coverage. We did not measure the Jina reranker’s tokenizer coverage of Ol Chiki, so the six dense successes removed during reranking are not assigned a cause. The shuffled-reference control pairs answers with same-language gold answers that are short, averaging roughly 15–21 characters across languages in our audit; absolute gap scales therefore depend on the scorer and should be interpreted within scorer. The structural audit checks citation form rather than claim-level support.

Generation used the archived Ollama tag llama3.1:8b at temperature 0, but the archived artifacts do not preserve the exact Ollama server version, model digest or quantization, and no explicit generation seed was set. These limits affect exact regeneration of the original answers, although the released outputs, evaluation code, fixed sample keys, and downstream diagnostic provenance are available for audit.

Ethics Statement

This study evaluates an experimental system using public datasets and does not involve human subjects or personally identifying information. The principal risk is overinterpretation of automatic scores in languages where model and judge coverage is uneven. A high score should not be used to justify deployment when retrieval, grounding, and language-specific validity have not been checked. The Santali results describe failures of this dataset and model stack, not a property of Santali or its speakers. Native-speaker evaluation remains necessary before making claims about system quality in consequential applications.

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A Full Diagnostic Tables

This appendix reports the language-level results summarized in Section 4. Languages appear in the order of Table 1.

A.1 Context Ablation

Lang	Token F1			BERTScore F1			Hit@5
	Gold	Retr.	No-ctx	Gold	Retr.	No-ctx	
Bengali (bn)	0.155	0.177	0.049	0.845	0.846	0.846	0.796
Tamil (ta)	0.178	0.187	0.043	0.856	0.855	0.850	0.805
Telugu (te)	0.126	0.147	0.048	0.852	0.851	0.847	0.534
Hindi (hi)	0.152	0.186	0.066	0.854	0.852	0.849	0.817
Urdu (ur)	0.165	0.175	0.083	0.825	0.824	0.816	0.511
Marathi (mr)	0.168	0.140	0.063	0.845	0.838	0.839	0.525
Gujarati (gu)	0.151	0.205	0.052	0.854	0.859	0.850	0.917
Nepali (ne)	0.135	0.152	0.043	0.840	0.841	0.836	0.743
Santali (sat)	0.144	0.066	0.083	0.905	0.902	0.907	0.011
Maithili (mai)	0.160	0.143	0.054	0.840	0.836	0.836	0.522
Macro mean	0.153	0.158	0.058	0.852	0.850	0.848	

Table 6: Full context-ablation results. Token F1 uses the same Unicode-aware normalization in every condition. Hit@5 applies to the retrieved condition only.

A.2 Retrieval Localization and Representation Diagnostics

Lang	Gold title in index	Dense Hit@25	Final Hit@5
Bengali (bn)	88.09%	84.1%	79.6%
Tamil (ta)	98.15%	89.3%	80.5%
Telugu (te)	92.60%	61.0%	53.4%
Hindi (hi)	90.81%	85.2%	81.7%
Urdu (ur)	57.48%	54.9%	51.1%
Marathi (mr)	81.26%	57.0%	52.5%
Gujarati (gu)	97.73%	94.9%	91.7%
Nepali (ne)	83.68%	79.1%	74.3%
Santali (sat)	79.73%	1.7%	1.1%
Maithili (mai)	63.40%	55.5%	52.2%

Table 7: Retrieval localization by index presence, dense retrieval, and final reranking. Index presence is measured over unique gold titles. Hit@25 is computed on BGE-M3 output before reranking and Hit@5 on the final reranked passages, both at the query level.

Diagnostic	Santali	Other nine (range)
Mean pairwise query cosine	0.9599	0.2816–0.3140
Mean cosine to query centroid	0.9799	0.5314–0.5602
Query BGE-M3 <unk> rate	47.85%	0%

Table 8: Query-representation diagnostics for Santali against the range across the other nine languages. BGE-M3 and xlm-roberta-large share a tokenizer, so the <unk> rate applies to both.

A.3 Judge Diagnostics

Lang	Faithfulness		Context precision		Answer Relevancy		
	Gold	Retr.	Gold	Retr.	Gold	Retr.	No-ctx
Bengali (bn)	0.603 (198)	0.733 (200)	0.379 (199)	0.730 (199)	0.442 (200)	0.463 (200)	0.580 (200)
Tamil (ta)	0.590 (190)	0.637 (197)	0.470 (182)	0.720 (178)	0.417 (200)	0.416 (198)	0.545 (199)
Telugu (te)	0.373 (200)	0.512 (199)	0.396 (127)	0.597 (136)	0.374 (198)	0.414 (199)	0.589 (200)
Hindi (hi)	0.462 (200)	0.649 (200)	0.327 (199)	0.668 (194)	0.397 (200)	0.549 (200)	0.684 (200)
Urdu (ur)	0.613 (200)	0.653 (198)	0.516 (196)	0.702 (196)	0.480 (200)	0.519 (200)	0.591 (200)
Marathi (mr)	0.439 (199)	0.570 (199)	0.349 (197)	0.542 (187)	0.331 (199)	0.460 (197)	0.632 (199)
Gujarati (gu)	0.524 (197)	0.543 (199)	0.422 (187)	0.674 (186)	0.401 (200)	0.470 (200)	0.593 (200)
Nepali (ne)	0.559 (198)	0.683 (197)	0.451 (197)	0.716 (194)	0.451 (194)	0.530 (200)	0.629 (200)
Santali (sat)	0.339 (196)	0.292 (197)	0.357 (198)	0.148 (195)	0.031 (197)	0.038 (197)	0.039 (196)
Maithili (mai)	0.581 (199)	0.584 (200)	0.576 (195)	0.682 (195)	0.461 (200)	0.468 (200)	0.633 (199)
Macro mean	–	–	–	–	0.379 (1,988)	0.433 (1,991)	0.552 (1,993)

Table 9: RAGAS scores by context condition; each mean is followed by the number of valid judgments. Faithfulness and context precision take the supplied contexts as input and are undefined for the no-context condition. Santali Answer Relevancy is not interpreted as a calibrated score (Section 4.5).

Condition	Faithfulness	Context precision	Answer Relevancy
Gold	1,977/2,000	1,877/2,000	1,988/2,000
Retrieved	1,986/2,000	1,860/2,000	1,991/2,000
No context	–	–	1,993/2,000

Table 10: Valid judgments by condition and metric out of 2,000 attempted (200 per language). Dashes mark metrics undefined without context.

A.4 Corpus and Retrieval-Context Diagnostics

Lang	Code	Tier	Corpus	Queries	Gold titles	Retr. titles	Query chars	Gold ans. chars	Avg ctx chars	Med ctx chars
Bengali	bn	High	767,965	1000	991	3613	62.7	19.2	466.4	387.5
Tamil	ta	High	686,000	1000	971	3786	68.8	20.1	427.7	343.0
Telugu	te	High	635,000	1000	986	3790	59.6	17.9	419.0	375.5
Hindi	hi	High	541,822	1000	979	3421	64.7	19.6	461.2	370.0
Urdu	ur	Mid	433,595	1000	929	3868	47.8	15.4	312.9	169.0
Marathi	mr	Low	203,479	1000	934	3425	59.1	16.6	351.9	269.0
Gujarati	gu	Low	99,357	1000	880	2626	58.6	21.1	482.0	422.0
Nepali	ne	Low	83,598	1000	925	2952	58.1	19.2	453.6	369.0
Santali	sat	X-low	43,500	1000	947	421	80.3	18.7	232.2	231.0
Maithili	mai	X-low	13,806	1000	918	2415	54.2	15.4	355.6	280.0

Table 11: Corpus and retrieval-context diagnostics. Corpus is the indexed passage count; retrieved-title counts and context lengths are computed from the saved reranked passages.

A.5 Structural Output Audit

Lang	Code	Empty	Short	Trunc.	No cite	Bad cite	Script leak	Med. chars
Bengali	bn	0.0%	0.0%	1.0%	1.0%	0.2%	0.6%	92
Tamil	ta	0.0%	0.0%	0.9%	9.4%	0.2%	1.0%	83
Telugu	te	0.0%	0.0%	0.8%	15.2%	0.5%	1.1%	78
Hindi	hi	0.0%	0.0%	0.1%	8.1%	0.5%	0.5%	104
Urdu	ur	0.0%	0.0%	0.1%	11.8%	0.3%	4.8%	80
Marathi	mr	0.0%	0.0%	0.0%	10.7%	1.2%	0.5%	75
Gujarati	gu	0.0%	0.0%	1.0%	22.5%	0.2%	0.7%	80
Nepali	ne	0.0%	0.0%	0.2%	3.2%	0.1%	0.6%	92
Santali	sat	0.0%	0.0%	1.1%	36.7%	0.2%	0.0%	64
Maithili	mai	0.0%	0.0%	0.2%	9.8%	1.9%	0.2%	81

Table 12: Structural audit of all 10,000 retrieved-condition generations ($n = 1000$ per language). *No cite* is citation omission; *bad cite* is a citation ID absent from the retrieved passages. The audit checks output form, not whether a cited passage supports the answer.